

Yogesh Pipada Sunil Kumar
The University of Adelaide
Adelaide, South Australia – 5000, Australia
yogeshpipada.sunilkumar@adelaide.edu.au
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Prof. Waldemar Rebizant
Editor-in-Chief
Electric Power Systems Research

Dear Prof. Rebizant,

Please find attached our manuscript entitled “*Comparative Evaluation of ReLU Neural Network Reformulations for Optimisation with Constraint Learning in Energy Aggregator Bidding*”, which we would like to submit for consideration for publication in *Electric Power Systems Research*.

This paper investigates tractable reformulations for embedding ReLU neural network surrogate models within optimisation problems for energy aggregator bidding. Within the optimisation with constraint learning (OCL) framework, neural networks are used to approximate nonlinear and decision-dependent prosumer flexibility costs. A controlled case study is employed to enable a systematic and fair comparison of different reformulation approaches.

The manuscript provides a comparative evaluation of several state-of-the-art neural network embedding reformulations, including mixed-integer formulations, penalty-based linear relaxations (PCAR and PCTAR), and an input convex neural network (ICNN)–based linear programming formulation. These approaches are assessed across multiple market price scenarios and neural network architectures with respect to computational tractability, surrogate accuracy, and realised optimisation performance.

The results reveal important methodological insights into the trade-offs between model expressiveness, computational scalability, and optimisation quality. In particular, the study demonstrates that predictive accuracy alone does not guarantee improved decision-making performance, and highlights the critical role of the interaction between surrogate models and their optimisation embeddings. The findings further show that structured architectures such as ICNNs can provide computationally efficient and reliable embeddings under suitable conditions.

We believe that this work is well aligned with the scope of *Electric Power Systems Research*, particularly in the areas of optimisation methods, data-driven modelling, and emerging techniques for flexibility aggregation and electricity market participation, thereby improving electrical energy utilisation.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree to its submission.

Thank you for your consideration. We look forward to your response.

Sincerely,
Yogesh Pipada Sunil Kumar
yogeshpipada.sunilkumar@adelaide.edu.au